Gas Planning Worksheet

Fill it in before every dive, for every diver on the team. ATA = (depth in feet ÷ 33) + 1. Tank factor = cubic feet per 100 psi.

1. What is my rock bottom?

The gas two divers need to solve a problem at depth and reach the surface or the next gas, at a stressed breathing rate.

_____ (ascent min, incl. 1 min to solve) × 1.0 (cu ft/min, stressed) × 2 (divers) × _____ (average ascent ATA) = _____ (cu ft)

_____ (cu ft) ÷ _____ (tank factor) × 100 = _____ (psi) → round up: _____ (rock bottom psi)

2. How much gas do I have?

Start pressure, as volume.

_____ (start psi) ÷ 100 × _____ (tank factor) = _____ (cu ft)

3. What is my usable gas?

Everything above rock bottom.

_____ (start psi) − _____ (rock bottom psi) = _____ (usable psi)

4. When do I turn?

All usable for a drift or live-boat pickup. Halves for a shore or anchored boat dive. Thirds when you must return to the start.

_____ (start psi) − (_____ (usable psi) ÷ _____ (1, 2 or 3)) = _____ (turn pressure psi)

5. Do I have enough gas for this dive?

Use your working SAC rate for the dive, then add rock bottom.

_____ (bottom time, min) × _____ (SAC, cu ft/min) × _____ (average ATA) = _____ (cu ft for the dive)

_____ (cu ft for the dive) + _____ (rock bottom cu ft) = _____ (cu ft needed) vs. _____ (cu ft I have (2))

6. How much bottom time does my gas give me?

_____ (usable cu ft) ÷ _____ (SAC, cu ft/min) ÷ _____ (average ATA) = _____ (minutes)

7. How fast will my gauge move at depth?

_____ (SAC, cu ft/min) × _____ (ATA) ÷ _____ (tank factor) × 100 = _____ (psi per minute)

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8. What is my SAC rate?

From a real dive: gas used, time and average depth.

_____ (psi used) ÷ 100 × _____ (tank factor) ÷ _____ (minutes) ÷ _____ (average ATA) = _____ (cu ft/min)

Team gas plan

Plan around the diver whose cylinder reaches rock bottom first.

Diver	Cylinder	Tank factor	Start psi	Rock bottom psi	Usable psi	Turn psi

Worked example: 60 feet on an aluminum 80

Problem at 60 ft. One minute to solve it, a minute to ascend to 30 ft, then one-minute stops at 30, 20 and 10 ft: 5 minutes, averaging about 33 ft, or 2.0 ATA.

5 (ascent min) × 1.0 (cu ft/min) × 2 (divers) × 2.0 (ATA) = 20 (cu ft)

20 (cu ft) ÷ 2.58 (tank factor) × 100 = 775 (psi) → round up: 800 (rock bottom)

3,000 (start psi) − 800 (rock bottom) = 2,200 (usable psi)

3,000 (start psi) − (2,200 (usable psi) ÷ 2 (halves)) = 1,900 (turn psi)

Based on UTD's gas planning method, using a stressed breathing rate of 1.0 cubic foot per minute per diver for rock bottom. For training and planning practice; learn and verify it with your instructor.

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